

DATA & OBSERVATION PATHWAY (per batch)

❄️ decoder layers 1...L — backbone θ frozen

Worked example

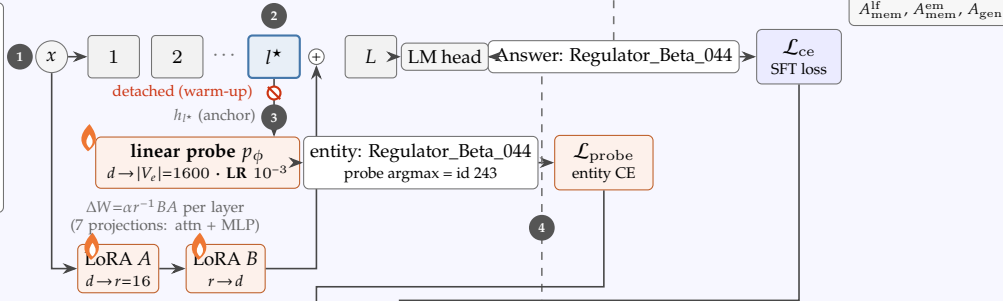
Q: Which regulatory authority oversees
Institution_Alpha_175 ?

A: Answer:

Regulator_Beta_044

probe label: entity id 243

anchor token: chars 36-57



CONTROL PATHWAY (per epoch)

saturation gate

$$\rho = \max(A_{\text{mem}}^{\text{lf}}, A_{\text{mem}}^{\text{em}})$$

$$\geq \tau_0 = 0.8 ?$$

closed: $\lambda = 0$
pure SFT

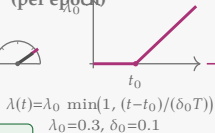
open at t_0 :
ramp $\lambda(t)$

worked example — 7B run

ep 24: $A_{\text{mem}}^{\text{lf}} = 0.805 \geq \tau_0$

gate opens; λ ramps to $\lambda_0 = 0.3$
(0.015 $\rightarrow$ 0.075 $\rightarrow$ 0.225)

0.5B: $\max A_{\text{mem}}^{\text{lf}} = 0.39$: never



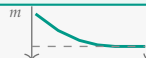
total objective

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \lambda(t) \mathcal{L}_{\text{probe}}$$

gradients

eval-aware controller

boost LR while
held-out improves;
decay $\times 0.7$ after degradation;
floor $m \leftarrow \max(0.5, m\gamma)$



LR $\times m(t)$

AdamW

updates
only A, B, ϕ
❄️ θ stays frozen

legend

- ❄️ frozen θ 🔥 trainable
- $\rightarrow$ eval-only signal
- ⊖ detached (no grad)